DRUGS · EAR PREPARATIONS

Aminoglycoside and combination ear drops

Aminoglycoside ear drops combine gentamicin, neomycin or framycetin, often with polymyxin and a steroid. They are effective, cheap and widely used for otitis externa. Their problem is the inner ear: if they reach the middle ear through a perforation or a ventilation tube, they can cause sensorineural loss and vestibular damage. Neomycin also sensitises the skin, and contact dermatitis can perpetuate the otitis externa it was meant to treat.

1 CHAPTER ONE At a glance

The drops on one page.

1.1 Summary

AMINOGLYCOSIDE AND COMBINATION EAR DROPS

	SUMMARY
Preparations	Gentamicin 0.3% with hydrocortisone 1%; neomycin with polymyxin B and hydrocortisone; neomycin with dexamethasone and acetic acid spray; framycetin combinations (availability varies)
How they work	Aminoglycoside protein-synthesis inhibition, polymyxin membrane disruption, steroid anti-inflammatory action
Spectrum	Pseudomonas and staphylococci (gentamicin); neomycin weak against Pseudomonas, polymyxin added for it; no antifungal activity
ENT uses	Acute otitis externa with an intact drum; eczematous otitis externa with superadded infection; granulation tissue
Dose	Typically 2 to 4 drops three or four times daily (sprays one spray three times daily) for 7 days
Avoid	Perforated drum or grommet where a safer alternative exists; known neomycin allergy
Pregnancy	Low systemic absorption from an intact canal

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Mechanism

Topical aminoglycosides reach very high concentrations in the canal. In the middle ear they diffuse through the **round window membrane** into the perilymph and damage hair cells, as systemic aminoglycosides do. The risk depends on the dose, the duration, and the state of the middle ear mucosa: inflamed, thickened mucosa may protect the round window, which is why toxicity is uncommon in actively discharging ears and more likely in a dry ear with a perforation.

3 CHAPTER THREE Using them in ENT

Indications and doses.

3.1 With an intact drum

They are a reasonable choice for **acute otitis externa** when the drum is seen and intact.

3.2 With a perforation or a tube

Guidance from ENT UK accepts that aminoglycoside drops may be used with caution in a **discharging** ear with a perforation when the benefit outweighs the risk, for **no longer than 2 weeks**, and with the patient warned; but quinolone drops avoid the dilemma and are preferred. Do not use them in a **dry** perforation.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Ototoxicity**: sensorineural hearing loss, tinnitus and vestibular failure if they reach the inner ear.
- **Contact dermatitis**: neomycin sensitivity is common in chronic otitis externa (patch tests are positive in a substantial proportion); a red, weeping, itchy canal that worsens on treatment suggests it.
- Otomycosis with prolonged use; steroid-related local effects.

4.2 Interactions

Clinically insignificant systemically. Avoid with other ototoxic ear preparations.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Mechanism of **topical ototoxicity** through the round window.
- The ENT UK position on aminoglycoside drops in a perforated ear.
- Neomycin contact sensitivity and its clinical picture.

5.2 Common pitfalls

COMMON PITFALLS

- Prescribing gentamicin drops for weeks in an ear with a perforation.
- Missing neomycin allergy in an otitis externa that gets worse with treatment.

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